

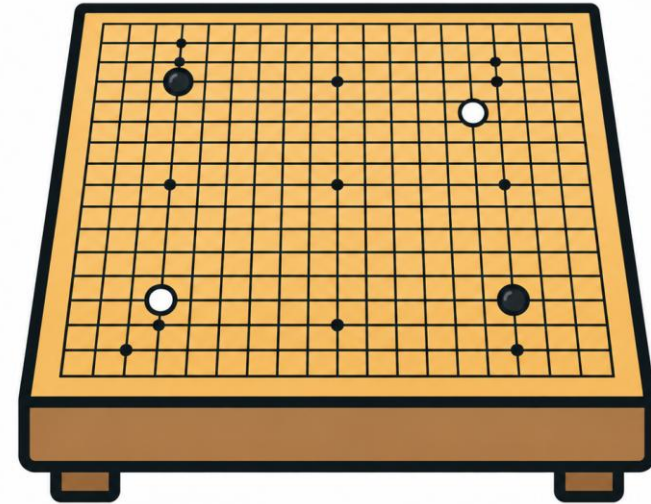
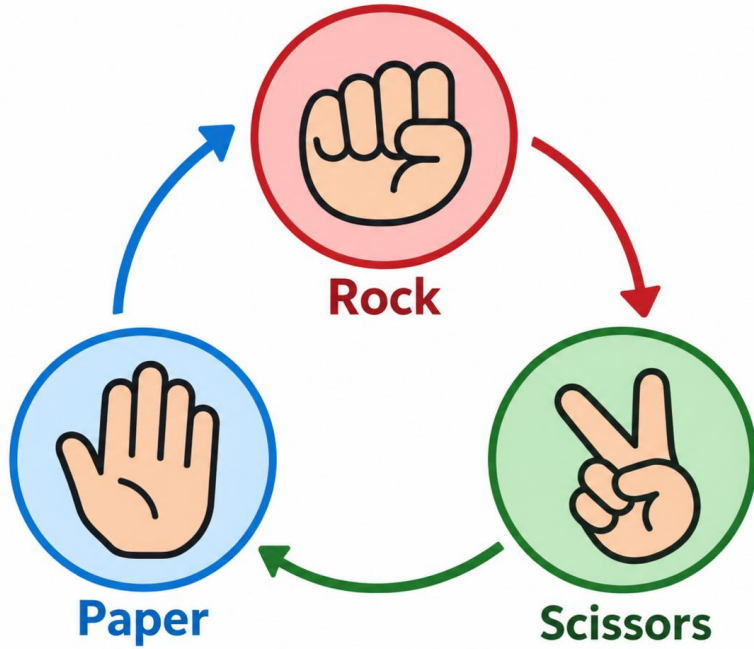
Two-Player Zero-Sum Games

Chen-Yu Wei

References

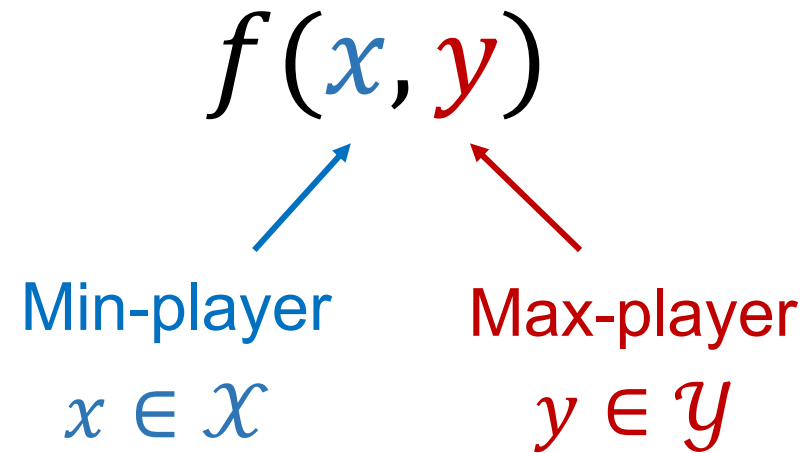
- Lecture 4 of [Haipeng's course](#)

Two-Player Zero-Sum Games



Go

Two-Player Zero-Sum Games



$$\mathcal{X} = \{ \text{Rock, paper, scissor} \}$$

$$\mathcal{X} = \Delta_3$$

Game Matrix

$$G = \begin{bmatrix} 1 & 1 & 0 & -3 \\ 2 & 3 & 1 & -5 \end{bmatrix}$$

cost matrix for row-player

Column player

Row player

			
	0	-1	1
	-1	0	1
	1	-1	0

If row-player uses $p \in \Delta_3$, column player uses $q \in \Delta_3$

$$\sum_{i=1}^3 \sum_{j=1}^3 p(i) q(j) G(i,j)$$

$$= p^T G q$$

$p \in \Delta_N, q \in \Delta_M$
 $G \in \mathbb{R}^{N \times M}$

Discussion (Sep 16)

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

$$\rightarrow \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Compare the Two Values

If we (min-player) have to play under this protocol, then by choosing

$$x^* = \operatorname{argmin}_x g(x) = \operatorname{argmin}_x \left(\max_y f(x, y) \right)$$

my cost is at most

$$\min_x \max_y f(x, y)$$

Proof,

$$\text{my cost} = f(x^*, \hat{y})$$

$$\leq \max_y f(x^*, y)$$

$$= \min_x \max_y f(x, y)$$

Max has to tell min their decision

their value $\geq \max_y \min_x f(x, y)$

$$\min_x \left(\max_y f(x, y) \right) \geq \max_y \left(\min_x f(x, y) \right)$$

depend on f

$g(x) = \text{my cost if I choose } x$

$h(y)$

→ Protocol: min-player has to tell max-player their decision

min $\hat{x}$ max $\hat{y}$

min $\hat{x}$
max $\hat{y}$

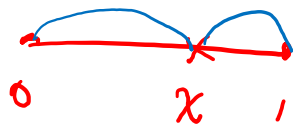
Compare the Two Values

$$\begin{aligned} \text{let } x^* &= \operatorname{argmin}_x \left(\max_y f(x, y) \right) \\ \min_{x \in X} \left(\max_{y \in Y} f(x, y) \right) &= \max_{y \in Y} \left(f(x^*, y) \right) \\ &\geq \max_{y \in Y} \left(\min_{x \in X} f(x, y) \right) \end{aligned}$$

<https://www.youtube.com/watch?v=RmgLPeASjXs>

Minimax Theorem Informal Illustration
(but this is actually not the minimax theorem...)

Compare the Two Values



Game values when both players do their best:

Sometimes this inequality is strict:

① $X = \{\text{rock, paper, scissor}\} = Y$

② $f(x, y) = |x - y|$, $X = Y = [0, 1]$

— $\uparrow$ convex in x & y

$$\min_x \max_y f(x, y)$$

$\geq$

$$\max_y \min_x f(x, y)$$

Protocol: min-player has to reveal their decision (x) to max-player

$$\min_{x \in [0, 1]} \left(\max_{y \in [0, 1]} |x - y| \right) = \min_{x \in [0, 1]} \left(\max \{x, 1 - x\} \right) = 0.5$$

Protocol: max-player has to reveal their decision (y) to min-player

$$\max_{y \in [0, 1]} \left(\min_{x \in [0, 1]} |x - y| \right) = \max_{y \in [0, 1]} (0) = 0$$

Minimax Theorem

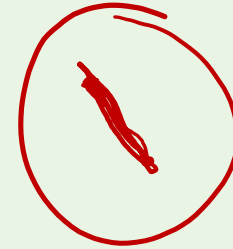
$$\boxed{X=Y=\Delta_3}$$

$$\sim \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_m$$

(John von Neumann's) Minimax Theorem (1928).

Let $G \in \mathbb{R}^{N \times M}$. Then

$$\min_{p \in \Delta_N} \max_{q \in \Delta_M} p^T G q = \max_{q \in \Delta_M} \min_{p \in \Delta_N} p^T G q$$



$$\min_{x \in X} \max_{y \in Y} f(x, y) = \max_{y \in Y} \min_{x \in X} f(x, y)$$

if X, Y are bounded convex sets, and $f(x, y)$ is convex in x , concave in y

$\forall y, \underline{f(x, y)}$ is a convex func of x

Discussion

Given a game matrix, (assuming the opponent will do their best), what is the best we can do?

$$\begin{aligned} p^* &= \underset{p}{\operatorname{argmin}} \left(\max_{\xi} p^T G \xi \right) \\ &= \underset{p}{\operatorname{argmin}} \left(\max_j (p^T G)_j \right) \end{aligned} \quad = \quad \begin{aligned} &\max_{\xi} \left(\min_p p^T G \xi \right) \\ &\max_{\xi} \left(\min_i (G \xi)_i \right) \end{aligned}$$

Proof by running algorithm

Δ_N : space of distribution over N actions

It suffices to prove

$$\min_{P \in \Delta_N} \max_{g \in \Delta_M} P^T G g \leq \max_{g \in \Delta_M} \min_{P \in \Delta_N} P^T G g$$

- ① Create an expert-learning problem
- ② run no-regret algorithm
- ③ get regret guarantee $\rightarrow$

$$P^T G g = \sum_{i=1}^N \sum_{j=1}^M P(i) g(j) G(i,j)$$

$$= \sum_{i=1}^N P(i) \left(\sum_{j=1}^M g(j) G(i,j) \right)$$

① For $t=1, 2, \dots, T$:

Generate $P_t \in \Delta_N$

Define loss $l_t = G g_t$ ($l_t = P_t^T G g_t$)

② For $t=1, 2, \dots, T$

Generate $g_t \in \Delta_M$

Define reward $r_t = G^T P_t$

$\langle g_t, r_t \rangle = P_t^T G g_t$

Regret for min-player

use exponential weights

$$\sum_{t=1}^T p_t^T l_t - \min_p \sum_{t=1}^T p^T l_t \leq 2\sqrt{T \log N}$$

$\parallel$
 $G g_t$

$$\Rightarrow \sum_{t=1}^T p_t^T G g_t - \min_p \sum_{t=1}^T p^T G g_t \leq 2\sqrt{T \log N}$$

$$\max_g \sum_{t=1}^T \underbrace{g^T r_t}_{G^T p_t} - \sum_{t=1}^T g_t^T r_t \leq 2\sqrt{T \log M}$$

$$\max_g \sum_{t=1}^T p_t^T G g - \sum_{t=1}^T p_t^T G g_t \leq 2\sqrt{T \log M}$$

$$\max_g \sum_{t=1}^T p_t^T G g - \min_p \sum_{t=1}^T p^T G g_t \leq 2\sqrt{T \log N} + 2\sqrt{T \log M}$$

$$\max_g \frac{1}{T} \sum_{t=1}^T p_t^T G g - \min_p \frac{1}{T} \sum_{t=1}^T p^T G g_t \leq \frac{2\sqrt{T \log N} + 2\sqrt{T \log M}}{T}$$

$$\bar{p} = \frac{1}{T} \sum_{t=1}^T p_t$$

$$\bar{g} = \frac{1}{T} \sum_{t=1}^T g_t$$

Contraction:

If $\min \max - \max \min = V_G > 0$
then let T large enough so that

$$\Rightarrow \max_g \bar{p}^T G g - \min_p p^T G \bar{g} \leq 0 \left(\sqrt{\frac{\log N}{T}} + \sqrt{\frac{\log M}{T}} \right) \quad \sqrt{\frac{\log M}{T}} \sqrt{\frac{\log N}{T}} < V_G$$

we know $\max_g \bar{p}^T G g \geq \min_p \max_g p^T G g$

$$\min_p p^T G \bar{g} \leq \max_g \min_p p^T G g$$

$$\Rightarrow \min_p \max_g p^T G g - \max_g \min_p p^T G g \leq 0 \left(\sqrt{\frac{\log N}{T}} + \sqrt{\frac{\log M}{T}} \right)$$

Nash Equilibrium

Another Proof of the Minimax Theorem

Application: Boosting

Setting: Binary classification

Question: Suppose we can always find a classifier that is only slightly better than random (i.e., 50% accuracy). Can we use this **weak learner** to produce an accurate classifier?

Setting

Training data: $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N) \in \mathcal{F} \times \{-1, 1\}$

Classifier: $h: \mathcal{F} \rightarrow \{-1, 1\}$

Loss of a classifier on a training sample:

What Does “Weak Learner” Mean?

Applying Minimax Theorem

From No-Regret Learning to Boosting

Input: Training dataset $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

Input: Classification oracle $\mathcal{O}$: weighted dataset $\rightarrow$ classifier

Initialize a no-regret algorithm $\mathcal{A}$ over Δ_N

For $t = 1, \dots, T$:

Let no-regret algorithm $\mathcal{A}$ output $p_t \in \Delta_N$ (distribution over data samples)

Let oracle $\mathcal{O}$ output a classification model h_t :

$$h_t = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{i \sim p_t} [1[h(x_i) \neq y_i]]$$

Define **reward** $r_t(i) = 1[h_t(x_i) \neq y_i]$.

Feed reward vector r_t to no-regret algorithm $\mathcal{A}$.

Caveat

- The discussion is only about the “training error”